

One Interval, Two Threshold Claims

Unit 3 · Topic 3.4 · THE PROBLEM

Suppose a statewide activities association has 10,000 member students and is considering requiring reusable food containers at its tournaments. In a simple random sample of 300 member students, 204 favor the requirement. A sponsor claims that at least 65% of all member students favor it; an event organizer makes the more modest claim that more than 60% favor it.

Kai: “The sample has $204/300 = 68\%$ in favor, above both thresholds, so it supports both claims.”

Mira: “The point estimate alone cannot settle both claims. We should construct a confidence interval and compare all of its plausible values with each threshold.”

Sample counts

Group	N	n	Favor	Other
Members	10,000	300	204	96

FIRST TAKE — one sentence before you work the parts

Do the data appear to support both claims, only one, or neither? Cite one number and commit before calculating the interval.

- DOK 1** (a) Define the population proportion p and calculate the sample proportion $\hat{p}$.
- DOK 2** (b) Verify the conditions, construct a 95 percent confidence interval for p , and interpret the interval in the association’s context.
- DOK 3** (c) In 3–4 sentences, decide which threshold claim the interval supports and whose reasoning is better. Compare the lower endpoint with 0.60 and 0.65. Explain why a plausible 0.65 is different from establishing at least 0.65.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.